

Table 1. HarmBench refusal-type analysis using the three-class FalseReject rubric. USR is toxic-prompt Useful Safety Rate: Direct Refusal plus Safe Partial Compliance. Direct and Adversarial regimes receive equal weight, so the five jailbreak templates do not outweigh Direct prompts. Per method, the 1B/3B rows contain 240 Direct and 1,200 Adversarial outputs; the compact 8B row contains 120 Direct and 600 Adversarial outputs. Values are full-sample percentages $\pm$ ten-group behavior-clustered jackknife standard errors.

Model	Method	Direct refusal (%)	Safe partial compliance (%)	Full compliance (%)	USR (%) $\uparrow$
Llama-3.2-1B-Instruct	Original	91.83 $\pm$ 1.03	0.92 $\pm$ 0.32	7.25 $\pm$ 1.21	<u>92.75 $\pm$ 1.21</u>
	A-LQR	90.00 $\pm$ 1.07	2.17 $\pm$ 0.83	7.83 $\pm$ 1.11	<u>92.17 $\pm$ 1.11</u>
	H_∞ (ours)	96.25 $\pm$ 0.78	0.88 $\pm$ 0.46	2.88 $\pm$ 0.68	97.12 $\pm$ 0.68
Llama-3.2-3B-Instruct	Original	82.33 $\pm$ 1.54	3.54 $\pm$ 1.03	14.12 $\pm$ 1.18	<u>85.88 $\pm$ 1.18</u>
	A-LQR	81.62 $\pm$ 1.79	4.58 $\pm$ 1.28	13.79 $\pm$ 1.32	<u>86.21 $\pm$ 1.32</u>
	H_∞ (ours)	91.79 $\pm$ 0.69	1.83 $\pm$ 0.63	6.38 $\pm$ 0.49	93.62 $\pm$ 0.49
Llama-3.1-8B-Instruct	Original	77.25 $\pm$ 3.40	5.58 $\pm$ 2.03	17.17 $\pm$ 2.61	<u>82.83 $\pm$ 2.61</u>
	A-LQR	80.17 $\pm$ 3.35	6.33 $\pm$ 1.46	13.50 $\pm$ 3.09	<u>86.50 $\pm$ 3.09</u>
	H_∞ (ours)	91.33 $\pm$ 1.81	2.33 $\pm$ 0.64	6.33 $\pm$ 1.36	93.67 $\pm$ 1.36